VULTR AGENT RUSH · BLAST RADIUS ZERO

Signed Scope Agent

The agent proposes a scope. A person approves it. Every run ends with a signed receipt and a destroyed sandbox.

Vultr Cloud Compute

Vultr Serverless Inference

APPROVED SCOPE

Domains	`acme-industrial.demo`
Methods	`GET`
Writes	`/work`
Caps	0.5 CPU · 256 MB · 20 s

```
BLOCKED collect.exfil-demo.net
CONTAINED rm -rf / · fork loop
SIGNED ed25519 · sandbox destroyed
```

THE PROBLEM

Agents that read the open web need a smaller blast radius

Hidden instructions

Supplier pages can carry text that only an AI agent will read, telling it to leak secrets or rank one vendor first.

Silent data leaks

A tracking pixel or script on the page can send data to a server nobody approved.

Hostile files

"Run our pricing calculator" is a normal request in procurement. The script can do anything.

No proof afterwards

When something goes wrong, nobody can show what the agent touched, wrote or tried to reach.

Buyer: vendor risk and procurement analysts who check supplier sites and price sheets every day.

HOW IT WORKS

Four steps, every run

1

Propose

A model on Vultr Serverless Inference proposes the narrowest scope: domains, methods, writable paths, CPU, memory and time. Code clamps it.

2

Approve

A person approves once. The scope is hashed. Anything outside it later needs a person again, through a scope-change card.

3

Contain

A fresh sandbox is built to exactly that scope. The only way out is a proxy that checks and logs every request.

4

Prove

Every process, file write and network attempt goes into an Ed25519-signed receipt. The sandbox is destroyed and the receipt says so.

THE DEMO

One vendor check, three containment moments

SUPPLIER	WHAT IT TRIES	WHAT STOPS IT
acme-industrial.demo	Points to a certificate registry outside the scope	Scope-change card. The agent waits for a person.
borealis-parts.demo	Hidden text asks agents to send API keys out. A pixel calls an outside server.	Injection flagged and ignored. Pixel blocked by the proxy.
cobalt-fasteners.demo	Its "pricing calculator" runs `rm -rf /` and a fork loop	No network, read-only root, 64-process cap, time cap. Clean retry.

Result: a supplier comparison where every value is quoted from its source, plus a receipt anyone can verify offline.

CONTAINMENT LAYERS

Each layer fails closed

Network

Per-run Docker network created with `--internal`. The scope proxy is the only exit. TLS tunnels refused. Code containers get `--network none`.

Filesystem

Read-only root. Writes only to a small tmpfs at `/work`. Inputs mounted read-only.

Process

All capabilities dropped, no-new-privileges, unprivileged user, CPU, memory and 64-process caps, wall-clock kill of the whole process group.

Lifetime

Every container and the network are removed at the end. The teardown is checked and the result is written into the signed receipt.

THE RECEIPT

Proof you do not have to take on trust

The receipt is canonical JSON: the task, the approved scope and its hash, every amendment, every event with its verdict, the results, the model used and whether the sandbox was destroyed.

It is signed with Ed25519. Change one verdict and the signature fails.

```
$ python verify.py receipt.json <pubkey>
OK: signature valid
run 91cfee0b2f scope 1fb233633ab4
events 41 blocked network attempts 1
sandbox destroyed True
```

HOW VULTR RUNS THIS

Built on Vultr end to end

Vultr Cloud Compute

Ubuntu 24.04 instance running the control app, the Docker engine that builds and destroys each sandbox, and Caddy for HTTPS. One cloud-init script sets it all up from the public repo.

Vultr Serverless Inference

Every model call: proposing the scope, screening pages for prompt injection, extracting terms with source quotes, and writing the analysis code that then runs in the sandbox. OpenAI-compatible API.

The model name is shown in the app and written into every receipt.

WHAT'S NEXT

From demo to daily use

Stronger isolation

Swap Docker for microVMs (Firecracker or gVisor) per run, with the same scope contract.

Real browsing

A headless browser behind a TLS-terminating scope proxy, so HTTPS supplier sites can be in scope.

Receipts that travel

Store receipts in Vultr Object Storage and attach them to the vendor file for audit.

Scope templates

Reusable, pre-approved scopes per team, so routine checks need one click.